

ADITYA KULKARNI

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EDUCATION

North Carolina State University

M.S. Computer Science

Raleigh, NC

Aug 2024 – May 2026

University of Mumbai

B.Tech. Computer Science, CGPA: 9.32/10

Mumbai, India

Aug 2020 – May 2024

WORK EXPERIENCE

Product Engineer Intern

May 2025 – Present

Isomer AI

Remote

- Delivered 10+ production REST API features for an AI-powered claims automation platform – Temporal.io workflows, RBAC, and real-time event pipelines – across 7 deployments using TypeScript, Node.js, Express, PostgreSQL, and Redis
- Integrated document processing pipelines (PDF generation, email parsing via Microsoft Graph API, S3 storage), Auth0 authentication, and multi-model AI tooling across 3 providers; contributed 394 commits and conducted code review throughout Agile sprint cycles

Graduate Research Assistant

Apr 2025 – Present

NC State University

Raleigh, NC

- Engineered a cloud-based genomic data platform on Google Cloud Run with JBrowse 2, enabling interactive visualization of 500GB of datasets for 100+ researchers worldwide
- Improved ML model accuracy 12% through hyperparameter tuning and time-series analysis on biological datasets

Software Developer Intern

Jun 2024 – Aug 2024

Mphasis (HP Company)

Remote

- Built 5+ AI-driven automation workflows including a health claim processing tool using OCR, NLP, and ML models, improving claim accuracy by 25% in production
- Automated CI/CD deployments on AWS and GCP, integrated Grafana monitoring dashboards, and supported model validation and red-teaming to enhance system reliability

PERSONAL PROJECTS

Inventory Reservation Service | Java 21, Spring Boot, PostgreSQL, Docker, GitHub Actions

2026

- Built a transactional inventory reservation REST API in Java 21 and Spring Boot preventing overselling via PostgreSQL row-level locking (SELECT FOR UPDATE) across multi-SKU reservations with idempotent endpoints and RFC 9457 error responses
- Verified concurrency correctness with Testcontainers – two concurrent requests against quantity 1 produce exactly one success and one 409 Conflict; Flyway migrations, Spring Security RBAC, multi-stage Docker build, GitHub Actions CI

PulseRank – Real-Time Leaderboard & Activity Feed | TypeScript, NestJS, MySQL, MongoDB, Redis

2026

- Built an event-driven leaderboard and activity feed platform – MySQL for transactional writes with idempotency-key deduplication and outbox pattern, MongoDB for feed document projections with TTL expiry, Redis sorted sets for sub-millisecond reads
- Validated multi-store consistency with Testcontainers integration tests, benchmarked hot paths with k6, and shipped CI/CD via GitHub Actions with OpenTelemetry and Prometheus observability

Expertiza – Ruby on Rails Open Source | Ruby, Rails, RSpec | 5K+ users, 300+ contributors

2024–2025

- Refactored core controllers in a production Ruby on Rails codebase applying SOLID and DRY principles; extracted business logic into service layers, reducing code redundancy by 35% and improving maintainability across 300+ global contributors
- Added 110+ RSpec tests covering happy path and edge cases across refactored controllers and APIs, increasing overall code coverage by 25%

TECHNICAL SKILLS

Languages: Java, Python, TypeScript, JavaScript, SQL, NoSQL, C, Ruby

Web & APIs: REST APIs, Node.js, Express, Spring Boot, NestJS, React, HTML, CSS

Databases: PostgreSQL, MySQL, MongoDB, Redis, Elasticsearch

DevOps & Cloud: AWS, GCP, Docker, Kubernetes, GitHub Actions, CI/CD, Terraform, Grafana

Frameworks & Tools: Temporal.io, Kafka, Spark, Hadoop, LangChain, PyTorch, TensorFlow